Tachy-Shield Edge AI Board

(Tachy-Pi-E13F1)

Hardware Datasheet

Board Rev. 1R00
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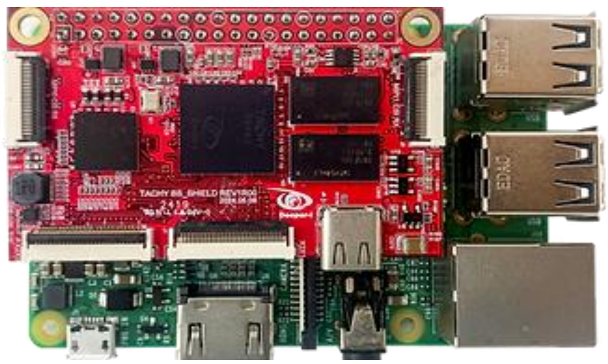
Reversion Sheet

Version	Date	Description
1.0.0	Jul. 2024	First Release
1.0.1	May. 2025	Tachy-Pi-E13F1 Overview image changed

Contents

1. **Tachy-Pi-E13F1** Overview

1.1 Introduction



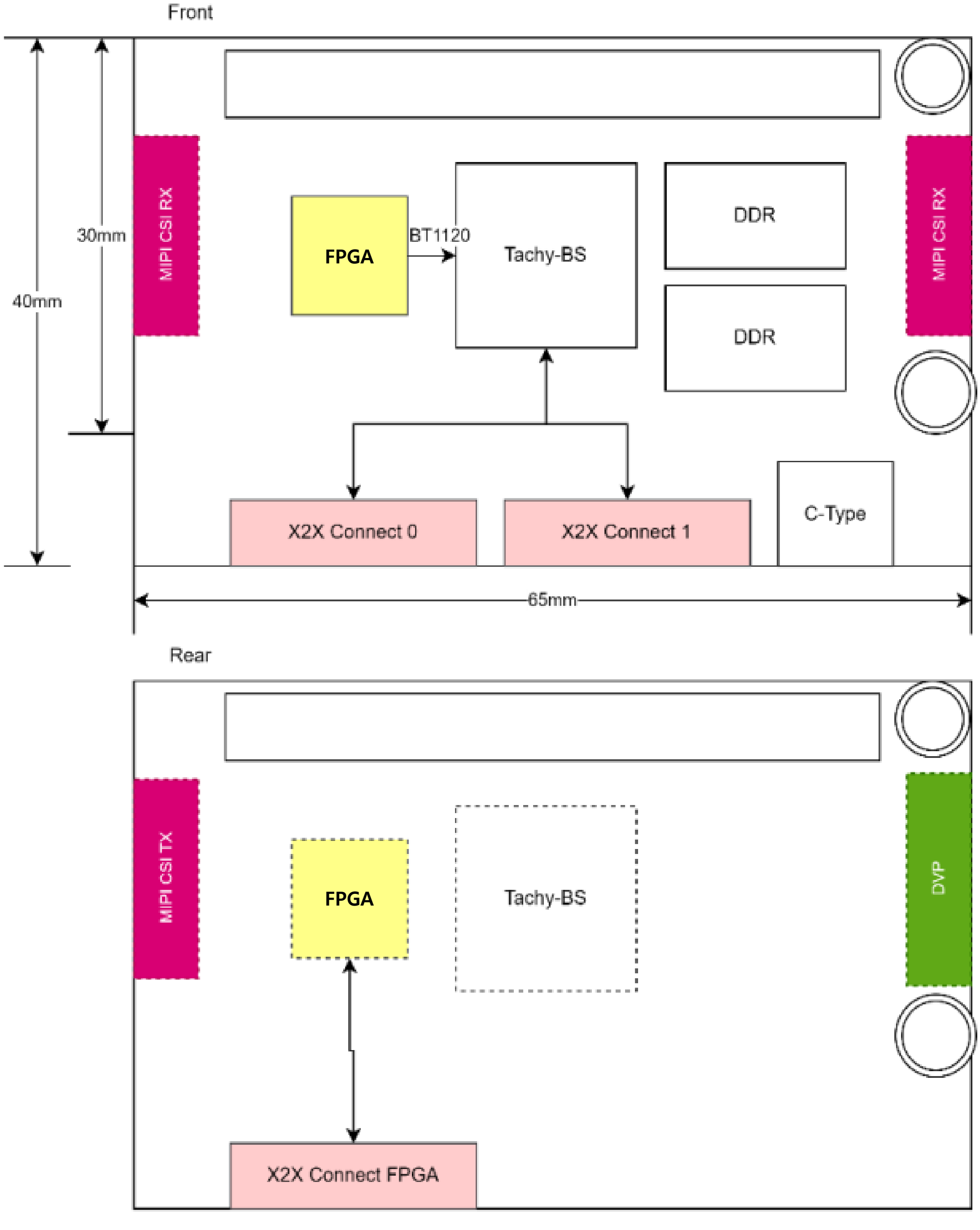
본 제품은 Raspberry Pi(A, B, Zero 등), Orange Pi 등 Pi류 SBC(Single Board Computer)와 호환 가능한 고성능, 저전력의 이미지 입력 딥러닝 연산 추론 가속 확장 보드로 다양한 카메라 입력과 칩간 통신 인터페이스를 통해 확장성, 데이터 전송 속도, 안정성을 보장합니다.

Deeper-I 고유의 고성능 NPU SoC Tachy BS402는 딥러닝 모델의 실시간 추론 가속 성능을 극대화하여 IoT, Edge Computing 환경 등 광범위한 응용 분야에 사용될 수 있습니다.

1.2 Hardware Features

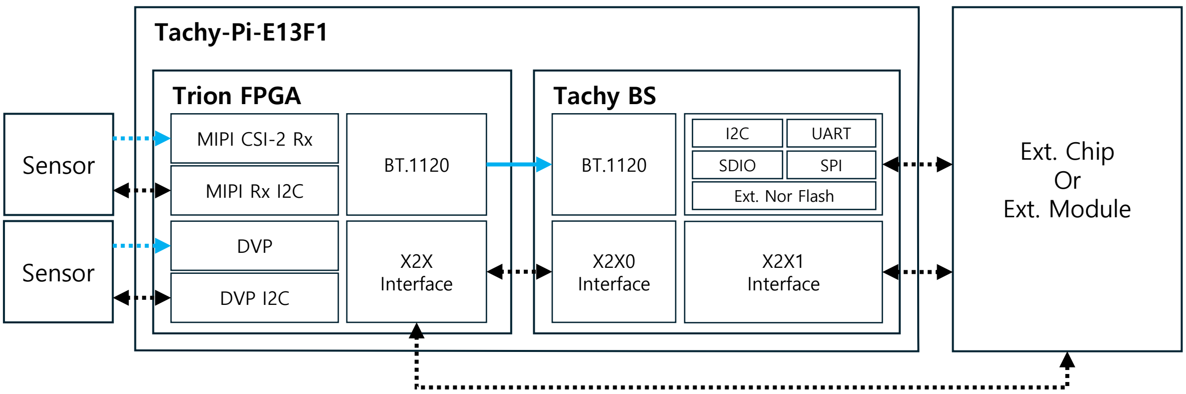
- Compatible with Raspberry Pi Series (A, B, Zero, etc.), Orange Pi, Banana Pi
- Stackable Pi SBC's 2.54mm 2 x 20-Pin Header
- Tachy BS NPU SoC Based (Cortex A5 CPU, Black Swan NPU)
- 1-GB DDR3 On-Board RAM
- Include FPGA for Camera Interface
- Up to 2 x MIPI CSI-2 RX
- Up to 1 x MIPI CSI-2 TX
- Up to 1 x DVP (In/Out)
- X2X Chip-to-Chip Interconnect
- 5 VDC (2.54mm 2 x 20-Pin Header Socket, Type-C USB Power Supply)
- 65 x 40 x 14 mm (W x H x L)

1.3 Board Layout



2. NPU Specification

2.1 Tachy BS(BlackSwan) NPU Introduction



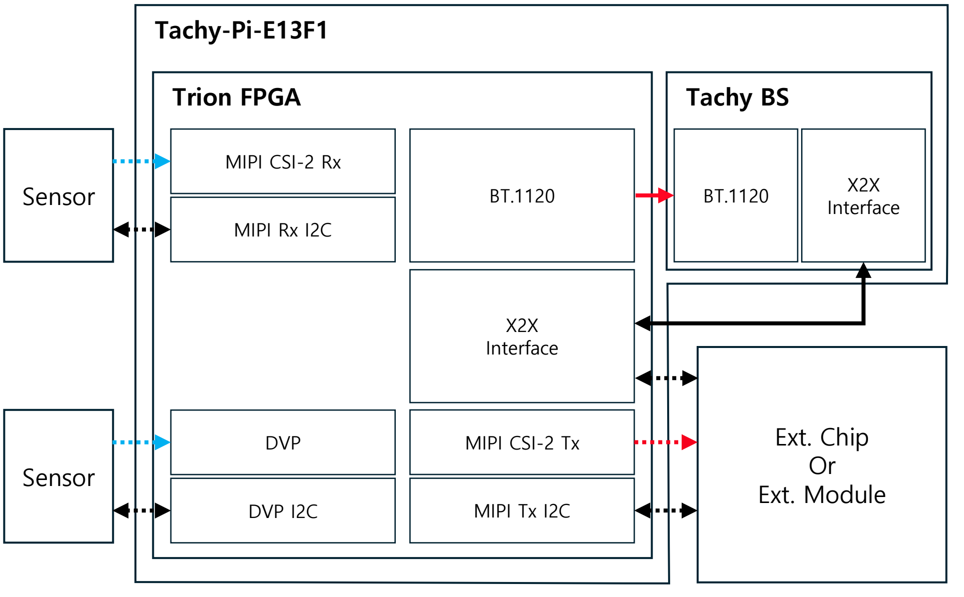
본 제품에 실장 된 Tachy-BS402는 Edge Device를 위해 고성능, 저전력으로 설계된 NPU SoC로 X2X Chip-to-Chip Interconnect 기술을 활용하여 병렬 분산 처리 시스템 구현이 가능하며 다양한 이미지 입력과 통신 Peripheral을 겸비하고 있습니다. 본 제품에서는 Trion FPGA로부터 BT.1120 입력을 받아 이를 처리하고 X2X Chip-to-Chip Interconnect와 Peripheral을 이용해 여타 칩, 모듈과 통신합니다.

2.2 Tachy BS(BlackSwan) NPU Specification

- ARM Cortex A5, 400-MHz Quad Core CPU
  - MMU, NEON, FPU Include
  - 32-KB L1 Cache (I/D), 128-KB L2 Cache
- 300-MHz Dual Core NPU
  - Low Power Deep-Learning Engine (Black Swan 1.0)
  - Network Optimized Operation, DMA Control
- (External) DDR3 533-MHz
- (External) NOR Flash
- Watch Dog Timer
- 3 x Timer (PIT) / PWM
- 2 x UART, 2 x SPI, 6 x I2C
- 120 x GPIO
- 1 x SDIO
- JTAG
- 1-Gbit Ethernet MAC (RGMII)
- 2 x X2C Chip-to-Chip Interconnect
- 2 x BT.1120 (Parallel 8-Bit Y, C), Bayer, RGB888
- FBGA-400
- 14 x 14 mm

3. FPGA Specification

3.1 Trion FPGA Introduction



본 제품에 실장 된 Tachy-BS402는 Edge Device를 위해 고성능, 저전력으로 설계된 NPU SoC로 X2X Chip-to-Chip Interconnect 기술을 활용하여 병렬 분산 처리 시스템 구현이 가능하며 다양한 이미지 입력과 통신 Peripheral을 겸비하고 있습니다. 본 제품에서는 Trion FPGA로부터 BT.1120 입력을 받아 이를 처리하고 X2X Chip-to-Chip Interconnect와 Peripheral을 이용해 여타 칩, 모듈과 통신합니다.

3.2 Trion FPGA(Trion TxxF169) Specification

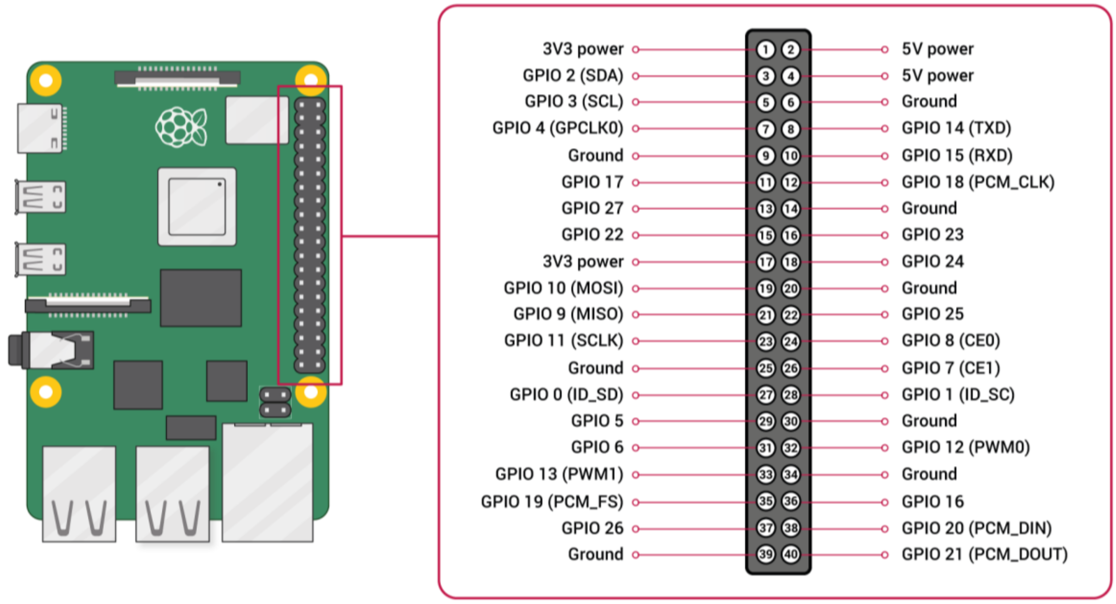
- Quantum Architecture (SMIC 40nm)
- 13~20K Logic Elements
- 727.04-Kbits Block RAM
- 5 x PLL Block (Max PLL Clock Output : 13)
- 31 x GPIO (+ 42 x LVDS as GPIO)
- 24 x Multipliers
- 800-Mbps LVDS Tx(8 Pair), LVDS Rx (12 Pair)
- 2 x MIPI DPHY, CSI-2 Rx Hard Controller IP (4-lane Data, 1-lane Clock)
- 2 x MIPI DPHY, CSI-2 Tx Hard Controller IP (4-lane Data, 1-lane Clock)
- FBGA169 Package
- 9 x 9 mm

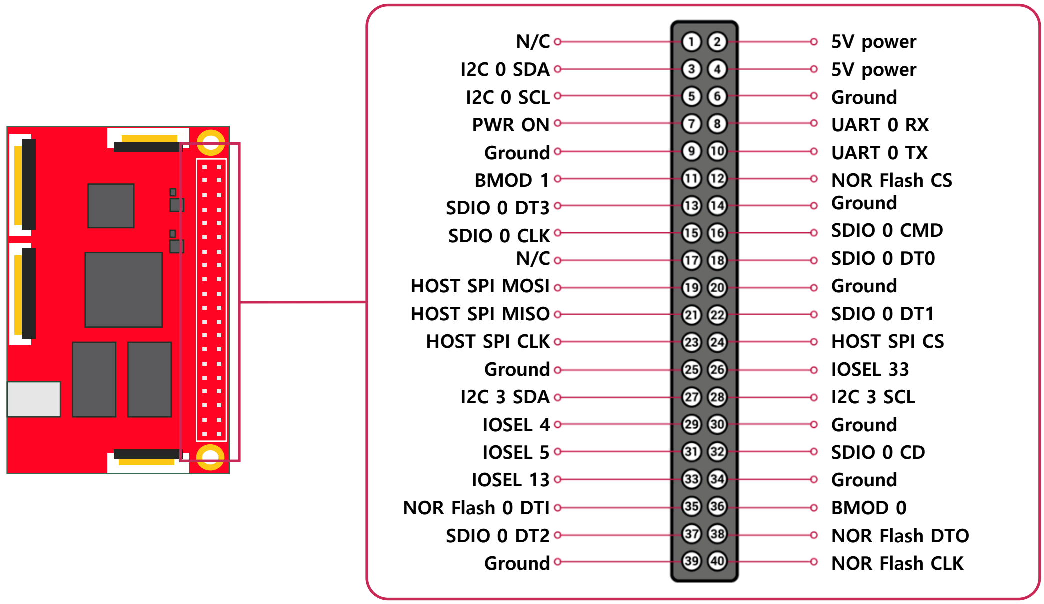
4. Board Interfaces

4.1 Pi HAT

4.1.1 2.54mm 2 x 20-Pin Header Socket

본 제품과 라즈베리 파이 간 연결은 아래 그림과 같이 2.54mm 2x20-핀 헤더를 이용해 구성됩니다. AI 연산을 위해 NPU는 주로 SPI 통신을 사용합니다.

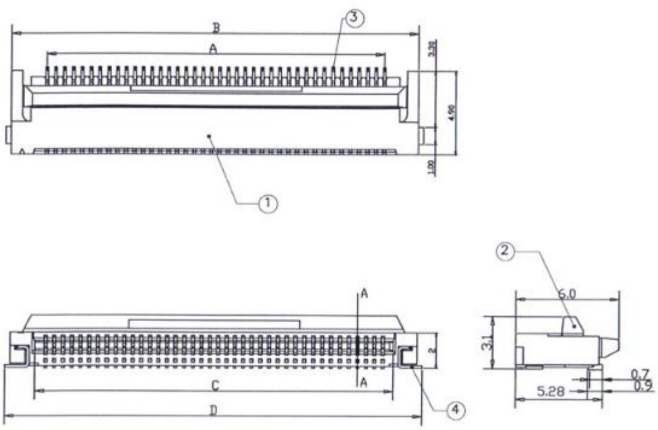




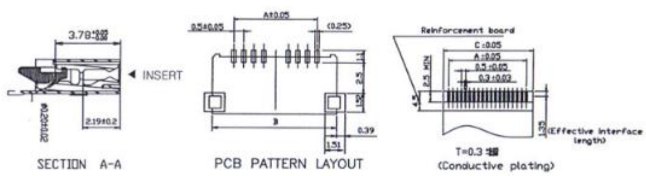
4.2 MIPI CSI-2 Rx=

4.2.1 22-Pin FPC Connector J4, J8

POS.	P/N.	A	B	C	D
22	TM0520-0022	10.5	14.5	11.5	15.1



4	LATCH	PHOS. BRZ	Sn(2~5)UM	2	
3	TERMINALS	PHOS. BRZ	CONTACT:AU	NN	
2	HANDLE	PPS+40GF	UL 94V-0	1	
1	HOUSING	LCP	UL 94V-0	1	
I/NO	DESCRIPTION	MATERIAL	FINISHED	Q'TY(unit:pcs)	REMARK

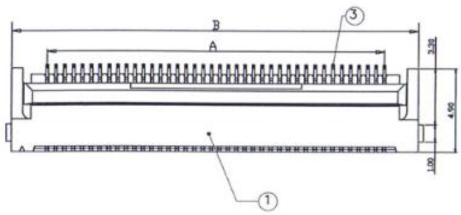


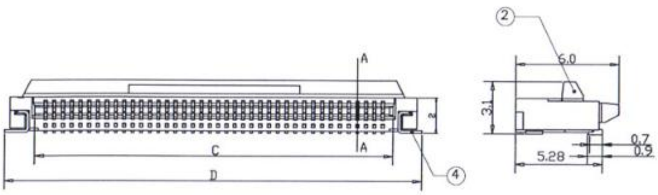
Pin Num	1	2	3	4	5
Desc.	GND	RX_DN0	RX_DP0	GND	RX_DN1
Pin Num	6	7	8	9	10
Desc.	RX_DP1	GND	RX_CLKN	RX_CLKP	GND
Pin Num	11	12	13	14	15
Desc.	RX_DN2	RX_DP2	GND	RX_DN3	RX_DP3
Pin Num	16	17	18	19	20
Desc.	GND	RX_GPIO0	RX_GPIO1	GND	RX_I2C_SCL
Pin Num	21	22			
Desc.	RX_I2C_SDA	3.3V			

4.3 MIPI CSI-2 Tx

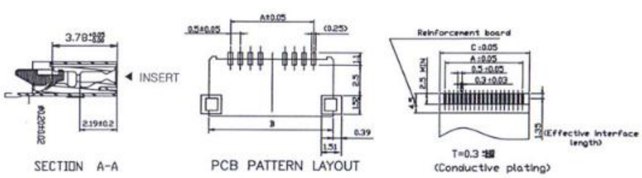
4.3.1 22-Pin FPC Connector J5

POS.	P/N.	A	B	C	D
22	TM0520-0022	10.5	14.5	11.5	15.1





4	LATCH	PHOS. BRZ	Sn(2~5)UM	2	
3	TERMINALS	PHOS. BRZ	CONTACT:AU	NN	
2	HANDLE	PPS+40GF	UL 94V-0	1	
1	HOUSING	LCP	UL 94V-0	1	
I/NO	DESCRIPTION	MATERIAL	FINISHED	Q'TY(unit:pcs)	REMARK

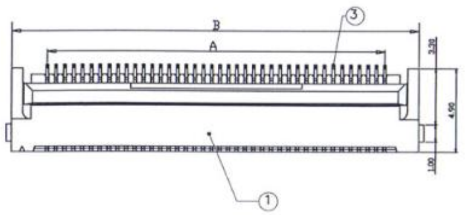


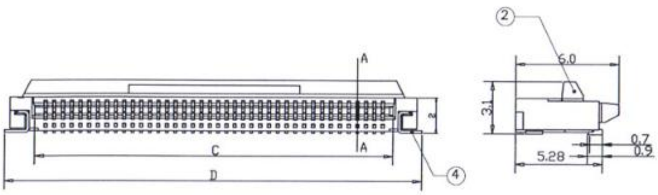
Pin Num	1	2	3	4	5
Desc.	GND	RX_DN0	RX_DP0	GND	RX_DN1
Pin Num	6	7	8	9	10
Desc.	RX_DP1	GND	RX_CLKN	RX_CLKP	GND
Pin Num	11	12	13	14	15
Desc.	RX_DN2	RX_DP2	GND	RX_DN3	RX_DP3
Pin Num	16	17	18	19	20
Desc.	GND	RX_GPIO0	RX_GPIO1	GND	RX_I2C_SCL
Pin Num	21	22			
Desc.	RX_I2C_SDA	GND			

4.4 DVP

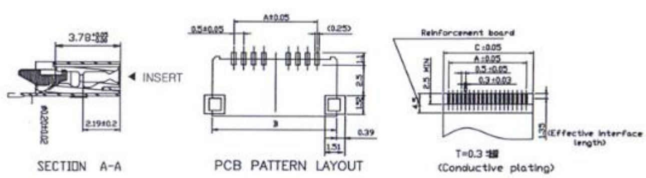
4.4.1 24-Pin FPC Connector J3

POS.	P/N.	A	B	C	D
24	TM0520-0024	11.5	15.5	12.5	16.1





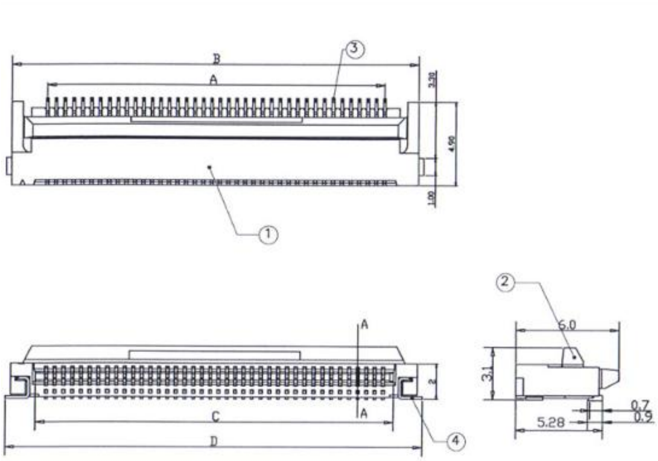
4	LATCH	PHOS. BRZ	Sn(2~5)UM	2	
3	TERMINALS	PHOS. BRZ	CONTACT:AU	NN	
2	HANDLE	PPS+40GF	UL 94V-0	1	
1	HOUSING	LCP	UL 94V-0	1	
I/NO	DESCRIPTION	MATERIAL	FINISHED	Q'TY(unit:pcs)	REMARK



Pin Num	1	2	3	4	5
Desc.	NC	GND	I2C_SDA	GND	SCL
Pin Num	6	7	8	9	10
Desc.	RST	VSYNC	PD	HSYNC	NC
Pin Num	11	12	13	14	15
Desc.	NC	D7	MCLK	D5	GND
Pin Num	16	17	18	19	20
Desc.	D5	PCLK	D4	D0	D3
Pin Num	21	22	23	24	
Desc.	D1	D2	GND	NC	

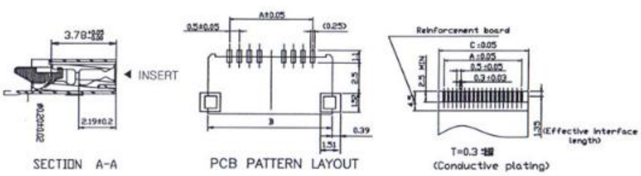
4.5 X2X Chip-to-Chip Interconnect

4.5.1 30-Pin FPC Connector J1 (for NPU SoC)



POS.	P/N.	A	B	C	D
04	TM0520-0004	1.5	5.5	2.5	6.1
06	TM0520-0006	2.5	6.5	3.5	7.1
10	TM0520-0010	4.5	8.5	5.5	9.1
12	TM0520-0012	5.5	9.5	6.5	10.1
15	TM0520-0015	7.0	11.0	8.0	11.6
16	TM0520-0016	7.5	11.5	8.5	12.1
20	TM0520-0020	9.5	13.5	10.5	14.1
26	TM0520-0026	12.5	16.5	13.5	17.1
28	TM0520-0028	13.5	17.5	14.5	18.1
30	TM0520-0030	14.5	18.5	15.5	19.1
36	TM0520-0036	17.5	21.5	18.5	22.1
40	TM0520-0040	19.5	23.5	20.5	24.1
45	TM0520-0045	22.0	26.0	23.0	26.6
50	TM0520-0050	24.5	28.5	25.5	29.1
54	TM0520-0054	26.5	30.5	27.5	31.1
60	TM0520-0060	29.5	33.5	30.5	34.1

4	LATCH	PHOS. BRZ	Sn(2~5)UM	2	
3	TERMINALS	PHOS. BRZ	CONTACT:AU	NN	
2	HANDLE	PPS+40GF	UL 94V-0	1	
1	HOUSING	LCP	UL 94V-0	1	
I/NO	DESCRIPTION	MATERIAL	FINISHED	Q'TY(unit:pcs)	REMARK



Pin Num	1	2	3	4	5
Desc.	GND	X2X1 PO CLK	GND	X2X1 TXVLD	X2X1 TXCMD
Pin Num	6	7	8	9	10
Desc.	X2X1 TXCHN0	X2X1 TXCHN1	X2X1 TXCHN2	X2X1 TXCHN3	X2X1 TXCHN4
Pin Num	11	12	13	14	15
Desc.	X2X1 TXRDY	X2X1 TXCH5	X2X1 TXCH6	X2X1 TXCH7	GND
Pin Num	16	17	18	19	20
Desc.	GND	X2X1 RXCHN7	X2X1 RXCHN6	X2X1 RXCHN5	X2X1 RXRDY
Pin Num	21	22	23	24	25
Desc.	X2X1 RXCHN4	X2X1 RXCHN3	X2X1 RXCHN2	X2X1 RXCHN1	X2X1 RXCHN0
Pin Num	26	27	28	29	30
Desc.	X2X1 RXCMD	X2X1 RXVLD	GND	X2X1 PI CLK	GND

4.5.2 30-Pin FPC Connector J1 (for NPU)

Pin Num	1	2	3	4	5
Desc.	GND	X2X0 PO CLK	GND	X2X0 TXVLD	X2X0 TXCMD
Pin Num	6	7	8	9	10
Desc.	X2X0 TXCHN0	X2X0 TXCHN1	X2X0 TXCHN2	X2X0 TXCHN3	X2X0 TXCHN4
Pin Num	11	12	13	14	15
Desc.	X2X0 TXRDY	X2X0 TXCH5	X2X0 TXCH6	X2X0 TXCH7	GND
Pin Num	16	17	18	19	20
Desc.	GND	X2X0 RXCHN7	X2X0 RXCHN6	X2X0 RXCHN5	X2X0 RXRDY
Pin Num	21	22	23	24	25
Desc.	X2X0 RXCHN4	X2X0 RXCHN3	X2X0 RXCHN2	X2X0 RXCHN1	X2X0 RXCHN0
Pin Num	26	27	28	29	30
Desc.	X2X0 RXCMD	X2X0 RXVLD	GND	X2X0 PI CLK	GND

4.5.3 30-Pin FPC Connector J7 (for FPGA)

Pin Num	1	2	3	4	5
Desc.	GND	X2XF PO CLK	GND	X2XF TXVLD	X2XF TXCMD
Pin Num	6	7	8	9	10
Desc.	X2XF TXCHN0	X2XF TXCHN1	X2XF TXCHN2	X2XF TXCHN3	X2XF TXCHN4
Pin Num	11	12	13	14	15
Desc.	X2XF TXRDY	X2XF TXCH5	X2XF TXCH6	X2XF TXCH7	GND
Pin Num	16	17	18	19	20
Desc.	GND	X2XF RXCHN7	X2XF RXCHN6	X2XF RXCHN5	X2XF RXRDY
Pin Num	21	22	23	24	25
Desc.	X2XF RXCHN4	X2XF RXCHN3	X2XF RXCHN2	X2XF RXCHN1	X2XF RXCHN0
Pin Num	26	27	28	29	30
Desc.	X2XF RXCMD	X2XF RXVLD	GND	X2XF PI CLK	GND